EX- 4: Water Jug Problem

Program:

```
from collections import deque

def water_jug():
    capacity_a = 4
    capacity_b = 3
    target = 2

    queue = deque([(0, 0, [])])
    visited = set()

    while queue:
        a, b, path = queue.popleft()

        if (a, b) in visited:
            continue

        visited.add((a, b))
        path = path + [(a, b)]

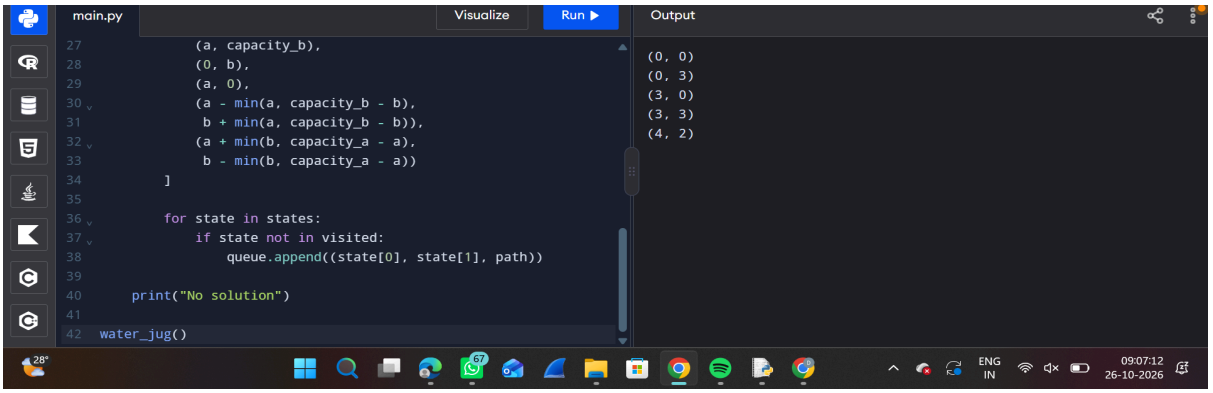
        if a == target or b == target:
            for state in path:
                print(state)
            return

        states = [
            (capacity_a, b),
            (a, capacity_b),
            (0, b),
            (a, 0),
            (a - min(a, capacity_b - b),
             b + min(a, capacity_b - b)),
            (a + min(b, capacity_a - a),
             b - min(b, capacity_a - a))
        ]

        for state in states:
            if state not in visited:
                queue.append((state[0], state[1], path))

    print("No solution")

water_jug()
```


main.py
Visualize
Run
Output
(a, capacity_b),
(0, b),
(a, 0),
(a - min(a, capacity_b - b),
b + min(a, capacity_b - b)),
(a + min(b, capacity_a - a),
b - min(b, capacity_a - a))
]
for state in states:
if state not in visited:
queue.append((state[0], state[1], path))
print("No solution")
water_jug()
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
ENG
IN
09:07:12
26-10-2026